

Tangents to polar curves

Key Points

For a polar curve $r(\theta)$:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta}(r\sin\theta)}{\frac{d}{d\theta}(r\cos\theta)}$$

To find horizontal tangents use

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(r\sin\theta) = 0$$

And for vertical tangents use

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(r\cos\theta) = 0$$

Basic Skills

Q1. Sketch the following polar curves for $0 \leq \theta < 2\pi$

a) $r = \cos 3\theta$ **b)** $r = 1$ **c)** $r = \theta$ **d)** $r = 2(1 + \cos \theta)$

Q2. Sketch $r = a \sec(\theta)$

Q3. Convert the coordinates

a) $(x, y) = (-2, -3)$ **b)** $(r, \theta) = (4, 5)$

Q4. Perform the following differentiations

a) $\frac{d}{d\theta}(\theta \sin \theta)$ **b)** $\frac{d}{d\theta}(\sin(\theta^2 + \pi))$

Competency Skills

Q5. For $r = \cos \theta$, $0 \leq \theta < 2\pi$

a) Find both $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$ and hence find θ such that $\frac{dy}{d\theta} = 0$ and $\frac{dx}{d\theta} = 0$

b) Hence state all the horizontal and vertical tangents (in cartesian form)

Q6. Find the tangents for the polar curve $r = 2(1 - \sin \theta)$, $0 \leq \theta < 2\pi$

Q7. Find the tangents for the polar curve $r = \sec \theta$

Q8. Find the area bounded inside the tangents of $r = \cos \theta$, $0 \leq \theta < 2\pi$

Advanced Skills

Q9. Find the area bounded outside of the curve $r = \cos \theta$, $0 \leq \theta < 2\pi$, but inside the rectangle formed by the curves tangents

Q10. At what coordinates do the tangents of $r = \sin \theta$, $-\pi \leq \theta < \pi$, intersect with the line $y = 2x - 1$

Q11. What is the cartesian gradient of the curve $r = \sqrt[2]{6 \cos \theta}$ when $\theta = \frac{\pi}{4}$

Q12. Find the stationary points, if they exist, of the curve $(x^2 + y^2)^{\frac{3}{2}} = 2xy$

Extension

Q13. Consider a particle moving in time whose position is described by polar coordinates $P(t) = (r(t), \theta(t))$, find the velocity $V(t)$